Extrema and the Mean Value Theorem

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Critical points and extrema

Find the critical points of $f(x) = 2x^3 - 3x^2 - 12x + 5$.

2 [Foundational] • Target: Critical points and extrema

Find the absolute maximum and minimum of $f(x) = x^3 - 3x$ on $[-2, 3]$, checking critical points and endpoints.

3 [Foundational] • Target: The Extreme Value Theorem

State the Extreme Value Theorem and explain why it guarantees $f(x) = x^2 - 4x$ has an absolute max and min on $[0, 5]$.

4 [Foundational] • Target: Rolle's Theorem

Verify that Rolle's Theorem applies to $f(x) = x^2 - 4x + 3$ on $[1, 3]$, and find the value(s) of c guaranteed by the theorem.

5 [Foundational] • Target: Rolle's Theorem

Explain why Rolle's Theorem does NOT apply to $f(x) = |x - 2|$ on $[0, 4]$, even though $f(0) = f(4) = 2$.

6 [Foundational] • Target: The Mean Value Theorem

For $f(x) = x^3$ on $[0, 2]$, find the value(s) of c guaranteed by the Mean Value Theorem.

7 [Foundational] • Target: The Mean Value Theorem

A car travels 180 km in 2 hours. Using the Mean Value Theorem, explain why the car must have been traveling at exactly 90 km/h at some instant.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Using the MVT to bound function values

Given $f(2) = 5$ and $-3 \le f'(x) \le 4$ for all x, use the MVT to find the range of possible values for $f(6)$.

9 [Intermediate] • Target: Consequences of the MVT: monotonicity

Explain, using the MVT, why $f'(x) > 0$ on an interval implies f is strictly increasing on that interval.

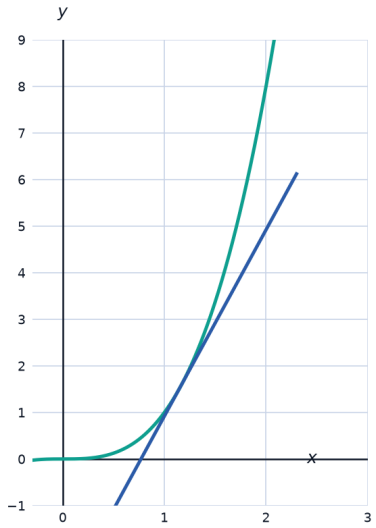
10 [Intermediate] • Target: Consequences of the MVT: monotonicity

Explain why two functions f and g with $f'(x) = g'(x)$ for all x must differ by a constant, using the MVT.

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11 [Intermediate] • Target: Visualizing the Mean Value Theorem

The graph shows $f(x) = x^3$ on $[0, 2]$ together with the secant line through the endpoints and the tangent line at $x = \frac{2}{\sqrt{3}}$ (found earlier).



12 [Intermediate] • Target: Absolute extrema on a closed interval — a capstone problem

Find the absolute maximum and minimum of $f(x) = x^4 - 4x^2 + 3$ on $[-2, 2]$, checking all critical points and endpoints.

13 [Intermediate] • Target: Rolle's Theorem with trigonometric functions

Verify that Rolle's Theorem applies to $f(x) = \sin x$ on $[0, \pi]$, and find the value of c guaranteed by the theorem.

14 [Intermediate] • Target: An applied Mean Value Theorem problem

A runner completes a 10 km race in 40 minutes. Using the Mean Value Theorem, find the runner's average speed, and explain why there must be some instant during the race when the runner's instantaneous speed equals this average.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Using the MVT to bound function values • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ for the function: $y = \int_{\cos x}^{x^3} \sqrt{1 - t^2}\, dt$.

16 [Challenging] • Target: Consequences of the MVT: monotonicity • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Level 1 · FoundationsFind the exact natural domain of the function $g(x) = \sqrt{\frac{x-1}{x+3}} - \ln(5 - x)$ in interval notation.

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17 [Challenging] • Target: Visualizing the Mean Value Theorem • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Level 1 · FoundationsFind $h'(2)$ given the accumulation function $h(x) = \int_0^{x^2} \frac{t}{t+1}\,dt$.

18 [Challenging] • Target: Absolute extrema on a closed interval — a capstone problem • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Level 3 · Advanced Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the boundary sphere $S : x^2 + y^2 + z^2 = 1$.

19 [Challenging] • Target: Rolle's Theorem with trigonometric functions • Cross-Lesson: 8.01 Area Between Curves

Level 2 · MediumEvaluate the trigonometric limit: $\lim_{x \to 0} \frac{1 - \cos(3x)}{x^2}$.

20 [Challenging] • Target: An applied Mean Value Theorem problem • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Level 2 · MediumSolve the transcendental equation analytically for x in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Critical points and extrema

Given $f(-3) = -2$ and $-4 \le f'(x) \le 6$ for all x, use the MVT to find the largest possible value of $f(3)$.

a. -26

b. 40

c. 34

d. 28

22 [Foundational] • Target: The Extreme Value Theorem

Find the larger critical point of $f(x) = x^3 - 9x^2 + 24x$.

a. 4

b. 2

c. 6

d. 8

23 [Foundational] • Target: Rolle's Theorem

Given $-3 \le f'(x) \le 2$ for all x, use the MVT to find the width of the range of possible values for $f(9) - f(2)$.

a. 31

b. 5

c. 42

d. 35

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24 [Foundational] • Target: The Mean Value Theorem

Does Rolle's Theorem guarantee a $c \in (-4, 6)$ with $f'(c) = 0$ for $f(x) = (x - 1)^2$ on $[-4, 6]$?

a. Yes, but only because f is increasing on the whole interval.

b. Yes — $f(-4) = f(6) = 25$, and f is a polynomial (continuous and differentiable everywhere), so Rolle's Theorem applies; here $c = 1$.

c. No — although $f(-4) = f(6)$, f is not differentiable on $(-4, 6)$, so Rolle's Theorem does not apply.

d. No — Rolle's Theorem never applies to functions of the form $(x - 1)^2$.

25 [Foundational] • Target: Using the MVT to bound function values

A continuous function on a closed, bounded interval $[a, b]$ is guaranteed (Extreme Value Theorem) to be:

a. bounded and to attain its max and min

b. monotonic

c. unbounded

d. differentiable

26 [Foundational] • Target: Consequences of the MVT: monotonicity

Global extrema on closed interval: Extreme Value Theorem guarantees max and min are attained. Find them by checking:

a. all critical points and endpoints

b. only critical points

c. only endpoints

d. only inflection points

27 [Foundational] • Target: Visualizing the Mean Value Theorem

Proof MVT: steps include Rolle's theorem applied to $h(x) = f(x) -$

a. $f'(a)(x - a)$

b. $f(a) - L(x)$ where L is the secant line

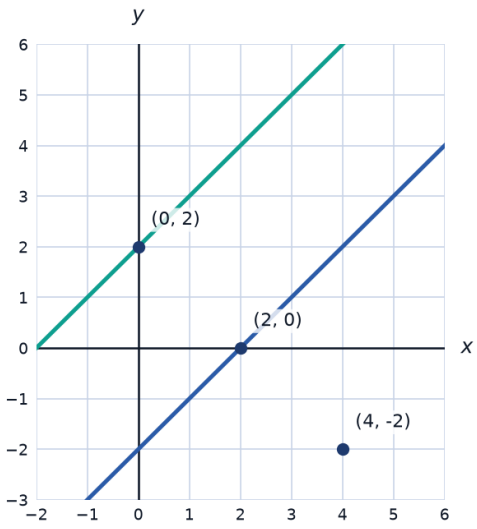
c. $f(b)(x - a)$

d. $f(a)$

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28 [Foundational] • Target: Absolute extrema on a closed interval — a capstone problem

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\,dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\,dx$?



a. 17

b. 7

c. 2.4

d. -7

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Rolle's Theorem with trigonometric functions**

The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx - 2$. The graph passes through the points $(1, -3)$ and $(2, 2)$. Determine the value of a.



a. 1

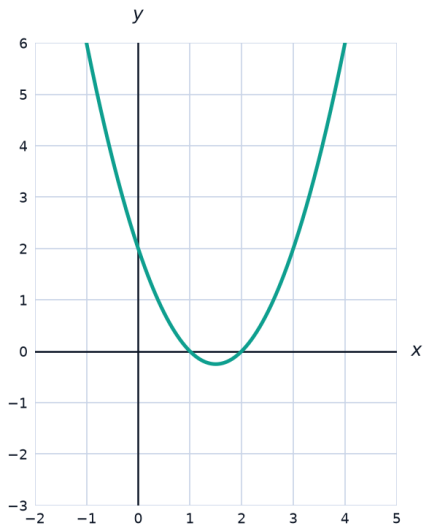
b. 2

c. 3

d. 4

30 **[Intermediate]** • **Target: An applied Mean Value Theorem problem**

Consider the function $f(x) = x^2 - 3x + 2$. The graph of $y = f(x)$ is a parabola. Find the x-coordinates of the points where the graph intersects the x-axis.



a. 1 and 2

b. -1 and -2

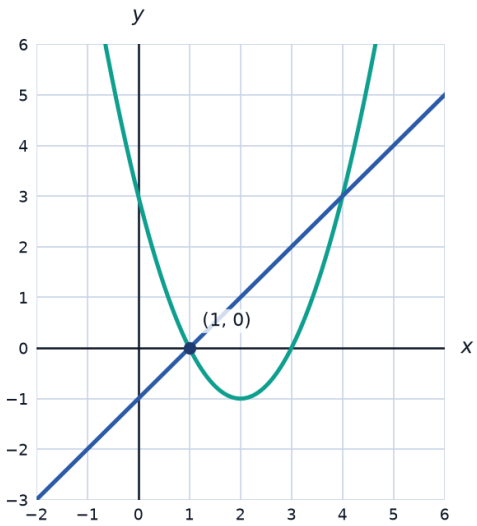
c. 2 and 3

d. -1 and 2

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31 **[Intermediate]** • **Target: Critical points and extrema**

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. The vertex of the parabola is at point C. Determine the coordinates of the point where the graph intersects the y-axis.



a. (0, 3)

b. (3, 0)

c. (0, -1)

d. (-1, 0)

32 **[Intermediate]** • **Target: The Extreme Value Theorem**

Consider the function $f(x) = \frac{ax^2 + bx + c}{dx + e}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = 3$. The graph also passes through the points $(0, 0)$ and $(1, -3)$. Find the value of $\frac{a}{d}$.

a. -6

b. -3

c. 3

d. 6

33 **[Intermediate]** • **Target: Rolle's Theorem**

Consider the function $f(x) = \frac{ax^2 + bx + c}{x - d}$, where $a, b, c, d \in \mathbb{R}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 3$ and a horizontal asymptote at $y = 2$. The function passes through the point $(1, -8)$. If $f'(x)$ has a critical point at $x = 5$, find the value of d.

a. 1

b. 2

c. 3

d. 4

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34 [Intermediate] • Target: The Mean Value Theorem

If $|f^{(4+1)}(c)| \le 1$ on the relevant interval, bound the Lagrange error $|R_4(x)|$ for approximating f at $x = 0.1$ using a Taylor polynomial centred at $a = 1$.

a. 0.0098415

b. 0.00492075

c. 0.00246038

d. 0.0273375

35 [Intermediate] • Target: Using the MVT to bound function values

Find the flux of $\mathbf{F} = \langle x, y, z \rangle$ through the closed unit sphere using the Divergence Theorem.

a. 2π

b. $\frac{4\pi}{3}$

c. 4π

d. $\frac{8\pi}{3}$

36 [Intermediate] • Target: Consequences of the MVT: monotonicity

Local extrema of f can occur only at critical points (interior) or on the:

a. axes

b. level curves

c. boundary

d. Hessian eigenvalues

37 [Intermediate] • Target: Visualizing the Mean Value Theorem

The divergence theorem applied to $\mathbf{F} = (x^3, y^3, z^3)$ over unit ball gives $\oiint_S \mathbf{F} \cdot \hat{n}\, dS = \iiint_V 3(x^2 + y^2 + z^2) dV =$

a. $4\pi/3$

b. 3π

c. $12\pi/5$

d. π

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Absolute extrema on a closed interval — a capstone problem • Cross-Lesson: 7.01 Differential Equations and Slope Fields

The method of Lagrange multipliers finds constrained extrema using multiplier λ. The critical value of λ itself:

a. is the rate of change of the optimal value w.r.t. the constraint constant

b. gives the extremum value

c. is always zero

d. equals $|ablaf|$

39 [Challenging] • Target: Rolle's Theorem with trigonometric functions • Cross-Lesson: 8.01 Area Between Curves

Mean Value Theorem guarantees c in open interval — this means the point is:

a. an endpoint

b. strictly interior

c. a maximum

d. undefined

40 [Challenging] • Target: An applied Mean Value Theorem problem • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Closed interval method: candidates for abs extrema are:

a. only critical points

b. only endpoints

c. critical points and endpoints

d. inflection points

41 [Challenging] • Target: Critical points and extrema • Cross-Lesson: 1.01 Limits Graphically and Numerically

Analytic function: uniqueness theorem — if f, g analytic on connected open set, and $f = g$ on a sequence of points converging to an interior point, then $f = g$:

a. on the boundary only

b. at all integers

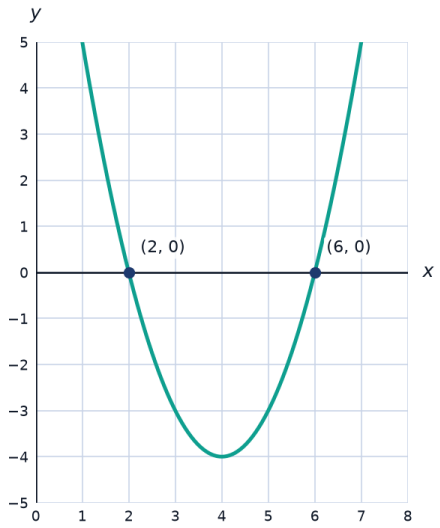
c. at a dense subset

d. everywhere on the domain

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42 [Challenging] • Target: The Extreme Value Theorem • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Consider the quadratic function $f(x) = x^2 - 8x + k$. The graph of $y = f(x)$ is a parabola that intersects the x-axis at the points $(2, 0)$ and $(6, 0)$. Find the value of k.



a. 12

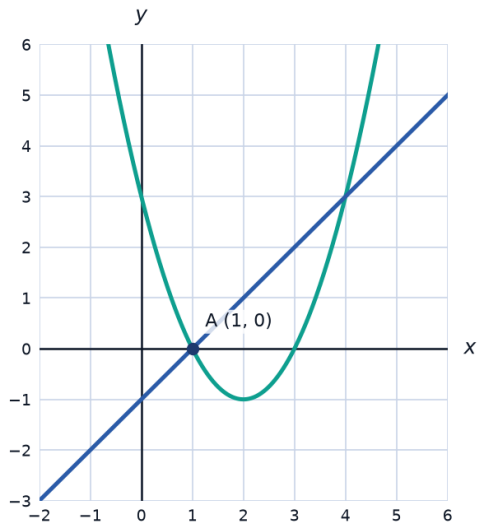
b. 16

c. 20

d. 24

43 [Challenging] • Target: Rolle's Theorem • Cross-Lesson: 2.01 The Definition of the Derivative

The diagram shows the graph of the function $f(x) = x^2 - 4x + 3$ and a line $y = g(x)$. The line $y = g(x)$ intersects the parabola $y = f(x)$ at points A and B. Point A is given as $(1, 0)$. Find the x-coordinate of point B.



a. 3

b. 4

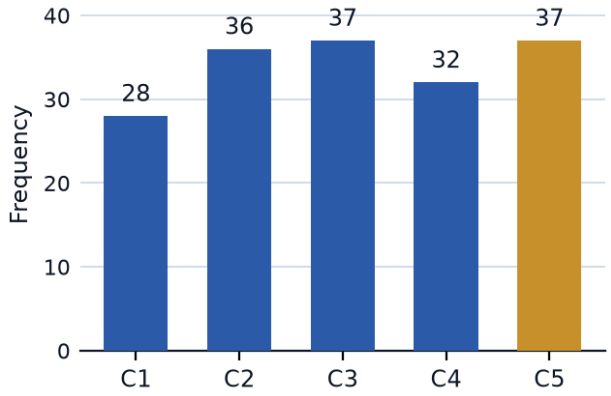
c. 5

d. 2

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44 [Challenging] • Target: The Mean Value Theorem • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

The diagram shows the graph of the function $y = f(x)$. The function is defined by $f(x) = x^2 + bx + c$. The graph passes through the point $(1, 0)$ and has a vertex at $(2, -1)$. Find the value of b.



a. -4

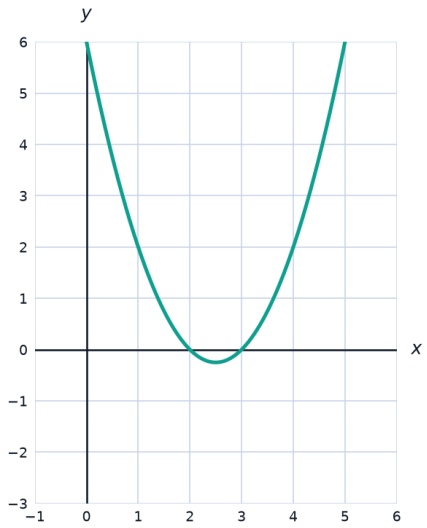
b. 4

c. -2

d. 2

45 [Challenging] • Target: Using the MVT to bound function values • Cross-Lesson: 3.01 The Chain Rule

Consider the function $f(x) = x^2 - 5x + 6$. The graph of this function is a parabola. Determine the x-coordinates of the points where the graph intersects the x-axis.



a. $1, 6$

b. $2, 3$

c. $1, 2$

d. $3, 4$

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